

Software Requirements Specification (SRS)

Smart Expense & Budget Management Application

Field	Detail
Project	Smart Expense & Budget Management Application
Course	Software Engineering – Mini Project (Section 5G)
Team	Team 14
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Team members

Sl. No	SRN	Name
1	PES2UG24CS427	Sahana Shivakumar Badiger
2	PES2UG24CS428	Sai Arjun M
3	PES2UG24CS429	Sai Arya A L
4	PES2UG24CS430	Sai Sathvik Gullipalli

Revision history

Version	Date	Author	Change summary	Approval
1.0	06-09-2026	Team 14	Initial SRS — requirements, use cases and RTM	

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			
Team Lead			

Table of Contents

1. Introduction 2. Overall description 3. External interfaces 4. System features 5. Non-functional requirements 6. Quality attributes & acceptance tests 7. UML use-case diagrams 8. Requirements Traceability Matrix

1. Introduction

1.1 Purpose

This document specifies the requirements for the Smart Expense & Budget Management Application, built by Team 14 for the Software Engineering mini project. It states what the application must do, how well it must do it, and how each requirement is verified. Readers: the project team, the course coordinator and the evaluators.

1.2 Scope

A web application in which one user records daily expenses and income, groups them into categories, sets a monthly budget, and sees where the money went through a dashboard with charts. In scope: user accounts, transaction entry and editing, categories, budgets with threshold alerts, dashboard and CSV export.

Out of scope: bank / UPI / card integration and statement import, shared or family budgets, native mobile apps, multi-currency and tax handling, and email or push notifications (alerts are shown in-app).

1.3 Audience

Team 14 developers, the course coordinator and the evaluation panel.

1.4 Definitions

Term	Meaning
SRS / RTM	Software Requirements Specification / Requirements Traceability Matrix
FR / NFR	Functional Requirement / Non-Functional Requirement
Transaction	A single expense or income record entered by the user

Term	Meaning
Category / Budget	A label grouping expenses / a spending limit set for a month
CSV / ORM	Comma-Separated Values file / Object-Relational Mapper
TLS / XSS	Transport Layer Security / Cross-Site Scripting

2. Overall description

2.1 Product perspective

A new, self-contained product with no external financial integrations. It uses a standard three-layer web architecture: a browser UI, a server-side application layer holding the business rules, and a relational database storing users, transactions, categories and budgets. All data comes from what the user types in.

2.2 Major product functions

- Register, log in and log out of a personal account
- Record, edit and delete expense and income entries under categories
- Set monthly and per-category budgets; get alerts at 80% and above 100%
- View monthly totals, a category breakdown chart and a six-month trend
- Filter and sort transactions and export the result as CSV

2.3 User roles

Role	Description and expectations
Registered User	Primary actor. Owns an account and all data in it. Basic web literacy; expects expense entry to take seconds and the dashboard to be self-explanatory.
Guest	Unauthenticated visitor. Can reach only the landing, registration and login pages.
Administrator	Team member who maintains the deployment, seeds default categories and takes backups, using server tooling rather than an admin screen.

2.4 Operating environment

Client: a modern browser (Chrome, Firefox or Edge) with JavaScript. Server: a web application server on Linux or Windows, deployed locally or on free-tier hosting. Data: a relational database (SQLite, or MySQL/PostgreSQL when deployed). Network: internet or campus LAN.

2.5 Constraints and assumptions

- Four students, one semester — the feature set is capped at the scope in 1.2
- Server-side web framework with a relational database; requirements are written to be framework-independent
- Only free and open-source libraries and free-tier hosting; a single currency (INR)
- No real bank, card or UPI credentials are handled anywhere in the system
- Users enter their own data honestly; the system verifies nothing against an external source

3. External interface requirements

3.1 User interfaces

A responsive web UI with a consistent header and navigation. Screens: Registration / Login; Dashboard (income, expense and balance cards, budget usage bar, category pie chart, six-month trend, alert banner); Transactions (table with filter, sort, edit and delete); Add / Edit Transaction form; Budget; Categories. Error messages appear next to the field they concern.

3.2 Hardware interfaces

None specific. The application runs on any desktop, laptop, tablet or phone that can run a supported browser; no peripherals are used.

3.3 Software interfaces

- Browser ↔ application server — HTTP over TLS; form posts and/or JSON endpoints
- Application server ↔ database — SQL through an ORM or parameterised queries
- Client charting library — renders the category and trend charts from aggregated JSON
- CSV export — server streams a CSV file with the correct content-type header

3.4 Communications interfaces

HTTPS (TLS 1.2+) in the deployed environment, with plain HTTP redirected. Session state in an HttpOnly, Secure cookie. Data exchanged as UTF-8 JSON. Failed requests offer a retry rather than discarding user input.

4. System features

Each requirement carries an acceptance criterion and a test case reference. IDs follow SEB-F-###.

4.1 User account and authentication

Req ID	Requirement (shall...)	Type	Priority	Source	Acceptance criteria / Test case
SEB-F-001	Allow a new user to register with a unique email, name and password.	Functional	High	End User	Unused email creates an account. TC-AUTH-01
SEB-F-002	Authenticate a user with email and password and start a session.	Functional	High	End User	Correct credentials open the dashboard; wrong ones show an error. TC-AUTH-02
SEB-F-003	Allow logout, after which protected pages are not reachable.	Functional	High	End User	After logout, /dashboard redirects to login. TC-AUTH-03
SEB-F-004	Restrict every user to their own records only.	Functional	High	Security	Requesting another user's record ID returns 403/404. TC-AUTH-04

4.2 Transaction and category management

Req ID	Requirement (shall...)	Type	Priority	Source	Acceptance criteria / Test case
SEB-F-005	Allow a user to add an expense with amount, date, category, payment mode and optional note.	Functional	High	End User	Saved expense appears in the list with the entered values. TC-EXP-01
SEB-F-006	Allow a user to record income with amount, date and source.	Functional	Medium	End User	Income is saved and counted in the monthly balance. TC-EXP-02
SEB-F-007	Validate that amount is a positive number and date is not in the future.	Functional	High	QA	Invalid input rejected with a field-level message. TC-EXP-03
SEB-F-008	Allow a user to edit an existing transaction.	Functional	High	End User	Edited values persist and totals recalculate. TC-EXP-04
SEB-F-009	Allow a user to delete a transaction after a confirmation prompt.	Functional	Medium	End User	Confirm deletes; cancel leaves the record. TC-EXP-05
SEB-F-010	Allow filtering and sorting of the transaction list by date range, category and amount.	Functional	Medium	End User	Only matching rows shown; total updates. TC-EXP-06
SEB-F-011	Provide default categories (Food, Travel, Rent, Utilities, Shopping, Health, Education, Other) and allow the user to add or rename their own.	Functional	Medium	End User	A user-created category is selectable on the expense form. TC-CAT-01

4.3 Budget management and alerts

Req ID	Requirement (shall...)	Type	Priority	Source	Acceptance criteria / Test case
SEB-F-012	Allow a user to set a monthly budget for the month and, optionally, per category.	Functional	High	End User	Saved budget is shown on the dashboard for that month. TC-BUD-01
SEB-F-013	Compute and display amount spent, amount remaining and percentage of budget used.	Functional	High	End User	Figures match the month's expense total within ₹1. TC-BUD-02
SEB-F-014	Show a warning alert at 80% of a budget and a distinct over-budget alert above 100%, without blocking new entries.	Functional	Medium	End User	Both alerts appear at the right thresholds; entry still allowed. TC-BUD-03

4.4 Dashboard, analytics and reports

Req ID	Requirement (shall...)	Type	Priority	Source	Acceptance criteria / Test case
SEB-F-015	Present a dashboard with total income, total expense and net balance for the selected month.	Functional	High	End User	Dashboard totals match the transaction list. TC-DSH-01
SEB-F-016	Display a category-wise pie chart and a six-month spending trend chart.	Functional	High	End User	Chart values match the underlying data; empty months show zero. TC-DSH-02
SEB-F-017	Allow export of the filtered transaction list for a date range as a CSV file.	Functional	Medium	End User	CSV downloads and row count matches the on-screen list. TC-RPT-01

5. Non-functional requirements

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
SEB-NF-001	Dashboard and transaction-list pages shall load within 3 seconds for an account holding up to 5,000 transactions.	Performance	High	Load time ≤ 3 s for 90% of requests on a 5,000-row seeded database. TC-PERF-01
SEB-NF-002	The application shall support at least 20 concurrent logged-in users without functional errors.	Scalability	Medium	20 simulated sessions complete with no 5xx errors. TC-PERF-02
SEB-NF-003	Committed data shall survive a server restart, and a failed save shall leave no partial record.	Reliability	High	Data consistent after restart; interrupted save rolls back. TC-REL-01
SEB-NF-004	A first-time user shall be able to record an expense within 3 clicks of the dashboard, with actionable error messages.	Usability	Medium	Walkthrough with 5 users; click count ≤ 3 . TC-UX-01
SEB-NF-005	The UI shall render correctly on the latest Chrome, Firefox and Edge, and be usable on a 360 px wide screen.	Portability	Medium	Manual check on 3 browsers and one mobile viewport; no clipping. TC-PORT-01

5.1 Security

5.1.1 Security objectives

SO-1 — Confidentiality: a user's transactions, budgets and account details must be readable only by that user, in transit and at rest.

SO-2 — Integrity and controlled access: records may be created or changed only by their owner through the application's validated paths, so stored balances always reflect what the owner entered.

5.1.2 Security requirements

Req ID	Requirement (shall...)	Priority	Acceptance criteria / Test case
SEB-SR-001	Carry all browser-server traffic over HTTPS (TLS 1.2+) in the deployed environment.	High	HTTP redirects to HTTPS; valid certificate. TC-SEC-01
SEB-SR-002	Store passwords only as salted one-way hashes (bcrypt / PBKDF2); never in reversible form or in logs.	High	Database holds only hashes with per-user salt. TC-SEC-02
SEB-SR-003	Authorise every request to a protected resource against the logged-in user's ID.	High	Tampering with a record ID returns 403/404. TC-SEC-03
SEB-SR-004	Use parameterised queries or an ORM and escape user text on output, to prevent SQL injection and stored XSS.	High	Injection payloads are stored and rendered as inert text. TC-SEC-04
SEB-SR-005	Enforce a minimum password length of 8 characters and lock login for 5 minutes after 5 consecutive failures.	Medium	Weak password rejected; 6th failed login blocked. TC-SEC-05

6. Quality attributes & acceptance tests

- Correctness — every total and chart value is recomputed from stored transactions and reconciled against the list during testing

- Usability — short forms, three-click expense entry, field-level errors, readable on a phone
- Security — hashed passwords, per-user authorisation, parameterised queries, HTTPS
- Performance and reliability — indexed queries on user and date; database transactions for multi-step writes
- Maintainability — modular code, README with setup steps, Git history with meaningful commits

6.1 Acceptance test suites

- TS-1 Authentication — TC-AUTH-01 to 04 TS-2 Transactions — TC-EXP-01 to 06, TC-CAT-01
- TS-3 Budgets — TC-BUD-01 to 03 TS-4 Dashboard and reports — TC-DSH-01, TC-DSH-02, TC-RPT-01
- TS-5 Non-functional — TC-PERF-01, TC-PERF-02, TC-REL-01, TC-UX-01, TC-PORT-01 TS-6 Security — TC-SEC-01 to 05

6.2 Exit criteria

All High-priority functional, non-functional and security requirements pass their acceptance criteria; every requirement in the RTM is mapped to an executed test case; no open Critical or Major defect at the final review.

7. UML use-case diagrams

7.1 Diagram 1 — Account, transactions and categories

Actors: Registered User (primary), Guest (limited), Database (supporting).

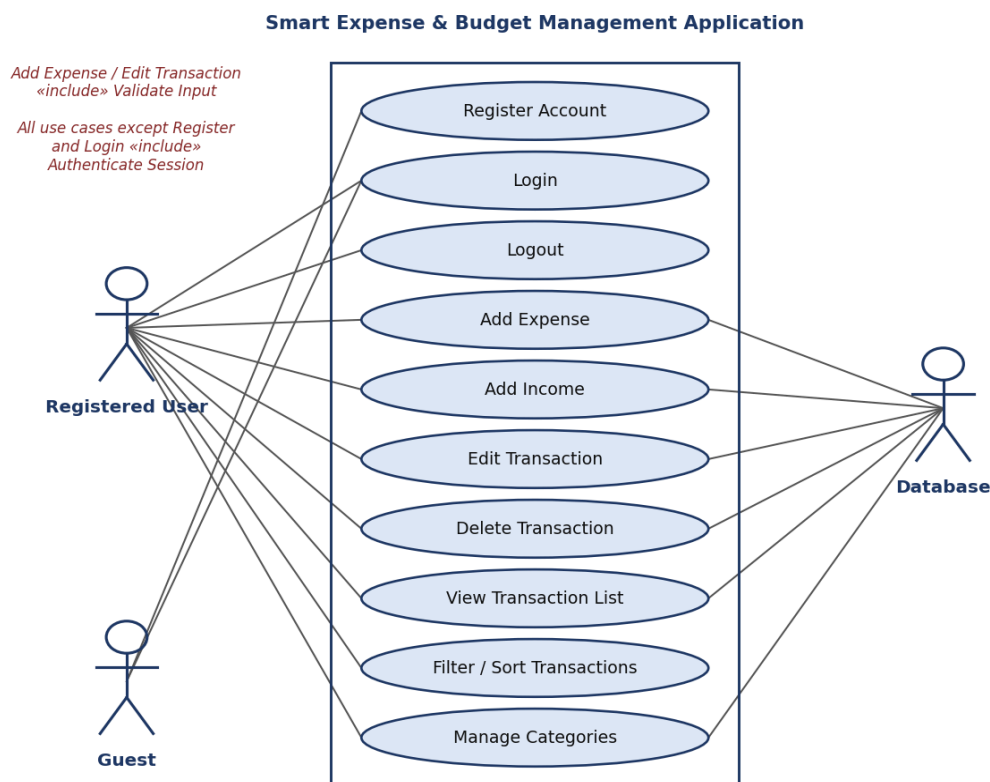


Figure 1 — Use-case diagram: account, transactions and categories

7.2 Diagram 2 — Budgets, dashboard and reports

Actors: Registered User (primary), Administrator (supporting), Database (supporting).

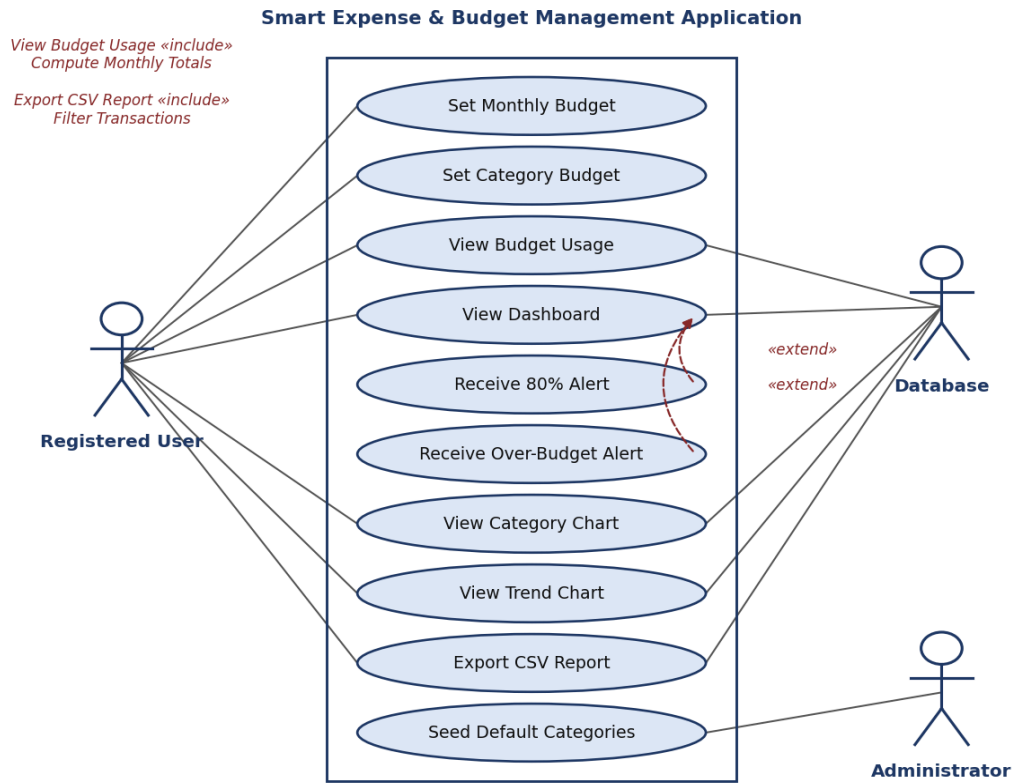


Figure 2 — Use-case diagram: budgets, dashboard and reports

8. Requirements Traceability Matrix (RTM)

Status: N = Not started, P = Partially implemented, A = Approved / verified.

Req ID	Requirement (short)	Section / Design spec	Module	Test case	Status
SEB-F-001	User registration	4.1 / DS-Auth-01	AuthModule	TC-AUTH-01	N
SEB-F-002	User login	4.1 / DS-Auth-02	AuthModule	TC-AUTH-02	N
SEB-F-003	Logout and session guard	4.1 / DS-Auth-03	AuthModule	TC-AUTH-03	N
SEB-F-004	Per-user data isolation	4.1 / DS-Auth-04	AuthModule	TC-AUTH-04	N
SEB-F-005	Add expense	4.2 / DS-Txn-01	TransactionModule	TC-EXP-01	N
SEB-F-006	Add income	4.2 / DS-Txn-02	TransactionModule	TC-EXP-02	N
SEB-F-007	Input validation	4.2 / DS-Txn-03	TransactionModule	TC-EXP-03	N
SEB-F-008	Edit transaction	4.2 / DS-Txn-04	TransactionModule	TC-EXP-04	N
SEB-F-009	Delete transaction	4.2 / DS-Txn-05	TransactionModule	TC-EXP-05	N
SEB-F-010	Filter and sort list	4.2 / DS-Txn-06	TransactionModule	TC-EXP-06	N
SEB-F-011	Manage categories	4.2 / DS-Cat-01	CategoryModule	TC-CAT-01	N
SEB-F-012	Set monthly budget	4.3 / DS-Bud-01	BudgetModule	TC-BUD-01	N
SEB-F-013	Budget usage figures	4.3 / DS-Bud-02	BudgetModule	TC-BUD-02	N
SEB-F-014	Budget alerts	4.3 / DS-Bud-03	BudgetModule	TC-BUD-03	N
SEB-F-015	Dashboard totals	4.4 / DS-Dsh-01	ReportModule	TC-DSH-01	N
SEB-F-016	Charts	4.4 / DS-Dsh-02	ReportModule	TC-DSH-02	N
SEB-F-017	CSV export	4.4 / DS-Rpt-01	ReportModule	TC-RPT-01	N
SEB-NF-001	Page load ≤ 3 s	5 / DS-Perf-01	WebUI / API	TC-PERF-01	N
SEB-NF-002	20 concurrent users	5 / DS-Perf-02	Server	TC-PERF-02	N
SEB-NF-003	Data durability	5 / DS-Rel-01	Database	TC-REL-01	N
SEB-NF-004	Ease of use	5 / DS-UX-01	WebUI	TC-UX-01	N
SEB-NF-005	Browser / mobile support	5 / DS-Port-01	WebUI	TC-PORT-01	N
SEB-SR-001	HTTPS everywhere	5.1.2 / DS-Sec-01	Deployment	TC-SEC-01	N

Req ID	Requirement (short)	Section / Design spec	Module	Test case	Status
SEB-SR-002	Hashed passwords	5.1.2 / DS-Sec-02	AuthModule	TC-SEC-02	N
SEB-SR-003	Authorisation checks	5.1.2 / DS-Sec-03	AuthModule	TC-SEC-03	N
SEB-SR-004	Injection / XSS defence	5.1.2 / DS-Sec-04	All	TC-SEC-04	N
SEB-SR-005	Password policy, logout	5.1.2 / DS-Sec-05	AuthModule	TC-SEC-05	N